A FINANCIAL AGENT FOR FUNDAMENTAL ANALYSIS: AN EMPIRICAL INVESTIGATION IN THE BRAZILIAN STOCK MARKET

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ABSTRACT

Large Language Models (LLMs) have recently enabled agentic systems for complex decision-making, including financial trading. However, existing Finance-AI research has primarily emphasized short-term signals such as sentiment and price prediction, leaving the ability of LLMs to perform Fundamental Analysis, a core, structured reasoning task, largely unexplored. In this work, we study whether LLMs can reliably compute fundamental financial indicators from real-world financial reports. We formulate Fundamental Analysis as a deterministic numerical reasoning task grounded in long-form financial documents, requiring the computation of 32 predefined indicators spanning accounting, valuation, profitability, leverage, and efficiency. We evaluate reasoning-enabled and non-reasoning LLMs at two model scales under workflow-based and agentic tool-augmented architectures, and analyze the impact of reflection mechanisms on numerical accuracy and robustness. Models are assessed using Normalized Mean Absolute Error (NMAE) and token-efficiency metrics. We further examine downstream decision quality by integrating model outputs into an LLM-based investment agent. Experiments on companies from the Brazilian stock market provide insights into LLM reasoning, grounding, and generalization beyond U.S.-centric settings.

1 INTRODUCTION

Financial trading plays a central role in modern markets, contributing to economic stability, efficient capital allocation, and effective risk management (Zhang et al., 2024). In this context, the volume of stream-level information available to support buy and sell decisions is enormous and readily accessible online. However, this information is continually growing and originates from heterogeneous sources and formats, such as financial news, tabular indicators, price charts, and earnings call recordings, making it cognitively infeasible for human traders to process and integrate all relevant signals without automated support (Yu et al., 2023).

More recently, Large Language Models (LLMs) have emerged as powerful tools for contextual understanding and generation in several applications, including finance. Early research in this domain has primarily explored the use of state-of-the-art LLMs to support human traders by automating operational and analytical tasks, such as summarization (Yang et al., 2024), sentiment analysis (Zhang et al., 2023a;b; Rodriguez Inserte et al., 2023), stock movement prediction (Yangjia et al., 2022; Tong et al., 2024; Liang et al., 2025) and question answering over financial documents (Islam et al., 2023; Han et al., 2024a). As LLM applications rapidly expand toward more complex decision-making and interactive settings (Ma et al., 2025), recent research in financial trading with LLMs has proposed agentic systems that autonomously collect financial data from multiple sources, distill key information, and leverage this information to execute buy or sell trading decisions or to construct investment portfolios (Yu et al., 2023; 2024; Zhang et al., 2024; Xiao et al., 2025).

Although recent research has significantly advanced the field, demonstrating that state-of-the-art AI models can simulate end-to-end trading companies (Fatouros et al., 2025), some key indicators traditionally used by human traders remain underexplored. In particular, prior work has largely overlooked Fundamental Analysis, which assesses a company's intrinsic value and long-term investment potential. Moreover, to the best of our knowledge, existing studies predominantly focus on the U.S. market, with limited attention to other financial contexts.

To address the gaps identified in prior work, this paper examines the extent to which LLMs can perform fundamental financial analysis in a controlled, reproducible setting. The task is to compute 32 predefined fundamental indicators across accounting, valuation, per-share, profitability, margin, leverage, and efficiency dimensions using the target company's quarterly and annual financial reports. The task is formulated as a deterministic numerical reasoning problem grounded in real-world financial documents, enabling direct comparison with reference values.

We evaluate both reasoning-enabled and non-reasoning LLMs at two model scales (nano and mini), under two architectural paradigms: (i) a workflow-based architecture, in which the set of financial reports for a given company is provided as input to the model; and (ii) an agent-based architecture, in which the model interacts with a structured database containing all available financial reports via SQL queries. Finally, we assess the impact of a reflection mechanism on both numerical accuracy and reasoning robustness.

Model versions were first evaluated by comparing automatically generated indicators with gold-standard indicators using an adapted version of the Normalized Mean Absolute Error (NMAE) metric. To compare the models' efficiency, we additionally report the average number of tokens consumed per analysis. To investigate the extent to which the best-performing approaches from the first assessment can support downstream investment decisions, we further input their generated reports into an LLM-based financial manager tasked with producing single-stock investment recommendations.

Unlike previous work, this study's empirical investigation was conducted with companies listed on the Brazilian Stock market to answer four research questions: **RQ1:** Can a Generative AI approach perform Fundamental Analysis of Stocks? **RQ2:** Do reasoning LLMs have a higher performance than non-reasoning ones? **RQ3:** Does the size of the LLM matter in the Fundamental Analysis? **RQ4:** Does an agentic LLM approach perform better than a workflow one? **RQ5:** Does a reflection mechanism leverage more accurate fundamental indicators? Our results indicate that the largest evaluated reasoning LLM (*GPT-5-mini*), implemented as a workflow without the reflection mechanist produces the best fundamental indicators, which proves to be helpful in support better investment decisions in financial trading simulations. This way, we contribute with empirical evidence on the applicability of generative and agentic LLMs to fundamental analysis in emerging markets, as well as with comparative insights about reasoning, scale, workflow design, and reflection mechanisms for financial decision-making.

2 Related Work

Early Generative AI research in finance focused on domain-specialized LLMs. Examples are BloombergGPT (Wu et al., 2023), a proprietary LLM, and FinGPT (Liu et al., 2023), an open-source one. Together with general-purpose LLMs, applications of finance-based LLMs were initially investigated for operational tasks in the domain, such as summarization (Yang et al., 2024), sentiment analysis (Zhang et al., 2023a;b; Rodriguez Inserte et al., 2023), stock movement prediction (Yangjia et al., 2022; Tong et al., 2024; Liang et al., 2025), question answering (Islam et al., 2023; Han et al., 2024a), among others (Wang et al., 2025b; Deng et al., 2025; Wang et al., 2026; 2025a).

With the rapid advancement of LLM applications in complex decision-making scenarios, recent research has investigated LLMs as decision-makers in financial trading. Yu et al. (2023) proposed FinMem, a trading agent with human-aligned memory consisting of a profile, a memory, and a decision-making module. The profiling module is responsible for defining the professional background knowledge of the analyzed stock (e.g., trading sector, historical financial performance overview, etc.) and the risk inclination the agent should assume. The memory module is the source of key financial facts separated into short-term (Daily Market News Insights), medium-term (Company 10-Q Report Insights), and long-term (Company 10-K Report Insights and reflection documents) memories. Finally, the decision-making module integrates the stock's profile and the main retrieved facts to support an investment decision.

FinCon (Yu et al., 2024) is a multi-agent framework for financial trading consisting of several LLM-based financial analysts aiming to process multi-modal market information, including textual (Daily

Financial News, Daily Market Sentiment, Quarterly 10-Q reports, Annually 10-K Reports, Monthly Investment Reports), tabular (Daily Asset Prices and Financial Metrics), and audio data (Earning conference call audio transcript). Simulating the hierarchy of investment firms, financial analysts interact with an LLM-based manager via natural language, which can take trading decisions for a single stock and manage portfolios. To support better decision-making, the frameworks also leverage a Risk-Control Component that monitors daily market risk and updates systematic investment beliefs through self-critique. Similarly, FinAgent (Zhang et al., 2024) is a multimodal LLM-based financial trader, focused on single trading decisions for stocks and crypto, that can reason over Asset Daily Prices, Asset News, Visual Data, and Expert Guidance to make investment decisions (Buy, Hold, or Sell).

Also inspired by trading firms, TradingAgents (Xiao et al., 2025) is a stock trading framework that sets LLM-powered agents with specialized roles such as finance analysts for summarizing different sources of information, researchers to write buy- and sell-oriented reports, traders to suggest an investment decision based on the research analyses, risk analysts to manage the risk exposure of the suggested decision and a manager, responsible for taking the final decision based on the previous generated content.

Although research has pushed the frontier far enough to see state-of-the-art AI models simulating a trading company on their own, we noticed that some important indicators used by human traders are not fully explored in related work. One of these is the traditional Fundamental Analysis of a stock. Among the financial analysts developed from studies simulating an investment firm with LLMs, TradingAgents is the only framework that includes an LLM-based specialist in Fundamental Analysis. However, the analysis is treated in a generic way, not asking the LLM to compute the popular fundamental indicators.

To address this gap, this study zooms in on Fundamental Analysis to benchmark LLM performance on the task.

3 Fundamental Analysis Task

Fundamental Analyses usually assess a company's intrinsic value, identifying its long-term investment potential and whether its stocks are over- or undervalued (Han et al., 2024b). In this study, the analysis is defined as a computational task that computes 32 financial indicators from the quarterly and annual financial reports of a target company.

Accounting Data consists of Balance Sheet and Income Statement Indicators. The former are represented by Total Assets (the total assets, rights, and value that the company owns); Cash and Cash Equivalents (the amounts that the company has in cash, banks, and cash equivalents); Current Assets (total assets and rights that the company owns and can be converted into cash in the short term, usually in one year); Shareholders' Equity (the total assets and rights that the partners and controllers have in the company); Gross Debt (the total debt that the company has); and Net Debit (the total debt that the company has minus cash and cash equivalents). Income Statement Indicators consists of Net Revenue (the total revenue a company has minus taxes, discounts, and returns); EBIT (Earning Before Interest and Taxes); Net Profit (the profit a company has after deducting all expenses and taxes). These indicators are computed for the quarter and the year, totaling 6 income statement indicators.

Valuation Indicators includes Price Ratios, like P/E (price-to-earning), P/B (price-to-book), P/EBIT (price-to-EBIT), Price-to-Sales, i.e., the stock price divided by net income per share), Price-to-Assets, Price-to-Working Capital (the share price divided by the current Assets minus Current Liabilities per share.) and Price-to-Net Current Assets). Enterprise ratios are also good metrics to analyse company's valuation, such as EV/EBITDA (Enterprise Value to Earnings Before Interest, Taxes, Depreciation, and Amortization) and EV/ EBIT (Enterprise Value to Earnings Before Interest and Taxes.

Per-Share Indicators are EPS and BVPS indicators, which stand for the earnings and book value per share, respectively.

Profitability and Margin Indicators includes Gross Margin (gross profit divided by net revenue), EBIT Margin (ratio between EBIT and net revenue) and Net Margin (net profit to net revenue) as margin indicators. For profitability, our approach computes EBIT-to-Assets, ROIC (return on invested capital), calculated by dividing EBIT by (Assets - Suppliers - Cash), and ROE (return on equity), calculated by dividing Net Income by Equity.

Leverage and Efficiency Metrics assess the company's debt health and are represented by Current Ratio (e.g., net current assets divided by current liabilities), Gross Debt-to-Equity ratio, and Asset Turnover (e.g., the net revenue divided by total assets).

4 Fundamental Analyst

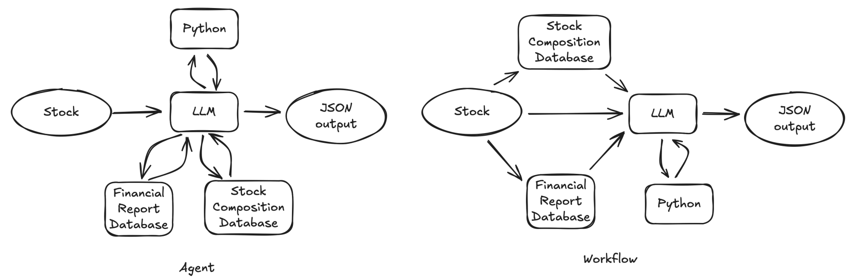

Figure 1: The proposed Fundamental Analyst designed as an Agent and as a Workflow.

This study proposes an LLM-based Fundamental Analyst that can compute 32 financial indicators from a target company's financial reports and its price over a given period.

Tools The proposed Fundamental Analyst has three tools at its disposal: an SQL database with financial reports of the companies and their number of common and preferred stocks of the companies, as well as a Python Code interpreter. More information about the databases can be found in the Appendix. Finally, the Python Interpreter tool aims to help the LLM analyst perform math operations and compute the Fundamental Indicators by writing Python code.

Structured Output The LLM of the proposed approach was developed to return the 32 financial indicators in a structured output that conforms to the JSON schema in the Appendix.

Agent vs. Workflow Anthropic (2024) distinguishes LLM-based agentic applications in Workflows and Agents, where the former orchestrates LLMs and tools through predefined code paths, and the latter has a less rigid structure, leveraging autonomy to allow an LLM to dynamically direct its own processes and tool usage. In this study, we explore both architectures for our Fundamental Analyst as depicted in Figure 1. In the workflow version of our approach, the set of financial reports for a given company is provided as input to the model, whereas in the agentic version, the model interacts with the structured databases as tools, being able to access the financial reports and stock composition via SQL queries.

Reflection Mechanism We also implemented a reflection mechanism at the end of the process that deterministically checks whether the approach's structured output contains the 32 financial indicators. If any indicators are missing, this mechanism calls the approach again, providing the missing indicators as feedback.

5 Experiment 1: Assessing Fundamental Indicators

In this first experiment, we aim to investigate how well the proposed financial analysts can compute fundamental indicators in contrast to gold-standard ones.

5.1 Data

We used data from 13 Brazilian publicly traded companies in the Energy Sector: Alupar Investimento (ALUP11), Auren Energia (AURE3), Companhia Paranaense de Energia (CPLE3), Engie Brasil Energia (EGIE3), Eletrobrás (ELET3), Eneva (ENEV3), Energisa (ENGI3), Equatorial

(EQLT3), ISA Energia Brasil (ISAE3), Light (LIGT3), Neoenergia (NEOE3), Renova Energia (RNEW11), and Serena Energia (SRNA3).

Balance Sheet data, Income Statements, and Fundamental indicators from these companies were extracted from the Fundamentus website[1] on April 17th, 2025. This information was used as the gold standard for automatic evaluation.

Every Brazilian company with stocks listed on the open market is required to submit quarterly and annual financial reports, similar to the 10-Q and 10-K reports of American companies, respectively. These reports are submitted to the *Comissão de Valores Mobiliários (CVM)*, an agency comparable to the American Securities and Exchange Commission (SEC). Brazilian quarterly reports are known as *Formulário de Informações Trimestrais (ITR)*, while the annual reports are called *Formulário de Demonstrações Financeiras Padronizadas (DFP)*.

In this study, we used the 2024 annual DFP forms for the 13 target companies as references. Both the annual forms and the number of outstanding stocks of these companies during the period were extracted from the CVM Open Data website[2]. The DFP annual data, structured as trees, were pruned to the third level of information and stored in a *sqlite* file, along with the number of stocks for each company.

5.2 Backbone LLM

Our approaches were built using LLMs from the GPT family. To investigate whether model size matters, we used the *mini* and *nano* versions of these models. In addition, to examine performance differences between reasoning and non-reasoning models, we employed GPT-5 and GPT-4.1, respectively. Overall, we evaluated approaches using four GPT-based LLMs: GPT-4.1-mini, GPT-4.1-nano, GPT-5-mini, and GPT-5-nano. For reasoning models (e.g., GPT-5), the reasoning and verbosity hyperparameters were set to *medium*. Both agent and workflow versions of the proposed approach were developed using the OpenAI Agents SDK[3].

5.3 Evaluation

We measure the error of our approach by contrasting the computed indicators with the gold-standard ones based on the following formula:

$$\mathrm{Score}(y, \hat{y}) = \mathrm{NMAE}(y, \hat{y}) + \alpha \cdot \mathrm{PenaltyRate}(y, \hat{y}) \tag{1}$$

The first part stands for the Normalized Mean Absolute Error (NMAE), defined as:

$$\mathrm{NMAE}(y, \hat{y}) = \frac{\frac{1}{N}\sum_{i=1}^{N} |y_i - \hat{y}_i|}{\max_{1 \le i \le N}(y_i) - \min_{1 \le i \le N}(y_i)} \tag{2}$$

where y represents the gold-standard values of the indicators, and $\hat{y}$ corresponds to the values generated by the agent. Since we noticed LLMs return 0s when they do not know how to compute a given indicator, the second part aims to penalize this number, as can be seen in the following equation:

$$\mathrm{PenaltyRate}(y, \hat{y}) = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left(\hat{y}_i = 0 \ \wedge \ y_i \neq 0\right) \tag{3}$$

where α is set to 10.

In addition to prediction quality, we also evaluated the efficiency of each approach by measuring the average number of consumed tokens per analysis. To ensure robust results, we report values based on the average of three runs for each approach.

[1] https://www.fundamentus.com.br/
[2] https://dados.cvm.gov.br/dataset/cia_aberta-doc-dfp
[3] https://openai.github.io/openai-agents-python/

5.4 RESULTS

Table 1: Results of Fundamental Analyst versions.

Model	Architecture	Reflect	NMAE	#Tokens
gpt-4.1-mini	agent	False	2.16	172502
	agent	True	1.57	195826
	workflow	False	∞	193569
	workflow	True	1.27	215343
gpt-4.1-nano	agent	False	70.2	85016
	agent	True	3.6	135225
	workflow	False	∞	159497
	workflow	True	1.19	224625
gpt-5-mini	agent	False	1	198611
	agent	True	1.01	218438
	workflow	False	0.43	204769
	workflow	True	0.63	246836
gpt-5-nano	agent	False	1.72	82059
	agent	True	0.66	65242
	workflow	False	∞	223107
	workflow	True	∞	260114

Table 5.4 depicts the average error for each approach. The first observation is that reasoning matters: GPT-5 outperforms all its counterparts, except GPT-5-nano, compared with GPT-4.1-nano in their workflow versions with reflection. Along the same lines, model size also matters, as *nano* models could not keep up with the performance of the *mini* versions in both reasoning and non-reasoning settings. Moreover, workflow-based architectures led to better results than the agentic approach. Finally, with respect to the reflection mechanism, it appears to yield better results for GPT-4.1 models and for the *nano* version of GPT-5, but not for GPT-5-mini. Overall, the configuration with the lowest error was GPT-5-mini using a workflow architecture without reflection.

Table 2: Comparison of indicator error scores across best workflow and agent approaches.

Term	Workflow 5-mini	Agent 5-nano Reflect	Term	Workflow 5-mini	Agent 5-nano Reflect
Accounting Data					
Total Assets	0	0.21	Current Assets	0	0.19
Cash and Equivalents	0.01	0.17	Gross Debt	0	0.38
Net Debt	0.01	0.52	Shareholders' Equity	0.01	0.15
Net Revenue (12 months)	0	0.28	Net Revenue (3 months)	0.04	0.35
Net Income (12 months)	0.02	0.16	Net Income (3 months)	0.02	0.27
EBIT (12 months)	0.09	0.25	EBIT (3 months)	0.07	0.38
Valuation Metrics					
P/E	0	0.10	P/B	0.08	0.47
P/EBIT	0.42	0.44	Price-to-Sales	0.17	0.47
Price-to-Assets	0.10	1.14	Price-to-Working Capital	0	0.12
Price-to-Net Current Assets	9.89	1.79	EV / EBIT	0.26	0.34
EV / EBITDA	0.08	0.11			
Per-Share Metrics					
EPS	0.04	0.33	BVPS	0.04	0.43
Profitability and Margin Metrics					
Gross Margin	0	0.24	EBIT Margin	0.09	0.20
Net Margin	0	0.12	EBIT / Assets	0.11	0.37
ROE	0.01	0.17	ROIC	0.11	0.38
Leverage and Efficiency Metrics					
Gross Debt / Equity	1.15	5.49	Asset Turnover	0.85	3.81
Current Ratio	0.06	1.24			

To better understand the error across individual indicators, Table 2 reports the error scores for each indicator obtained by the best workflow approach (GPT-5-mini workflow without reflection) and the best agent approach (GPT-5-nano agent with reflection). The results reinforce the pattern that workflow-based approaches perform better than agentic ones, with the former outperforming the latter across all indicators. When analyzing groups of indicators, we observe that the proposed approach shows greater facility in computing accounting and per-share indicators, which present the lowest error scores. In contrast, it appears to struggle more with valuation, profitability, and efficiency metrics, particularly those involving EBIT (P/EBIT, EV/EBIT, and ROIC), Price-to-Net Current Assets, Gross Debt / Equity, and Asset Turnover.

6 EXPERIMENT 2: ASSESSING INVESTMENT DECISIONS

In the second experiment, our goal is to evaluate whether the proposed financial analyst can help in leveraging good investment decisions and generate profit. For that, inspired by related work (Xiao et al., 2025), we simulate an investment firm in which the fundamental indicators computed by the proposed financial analyst are fed into an LLM-based manager that decides whether to sell, buy, or hold a target stock.

6.1 FINANCIAL MANAGER

The Financial Manager is based on *GPT-5-mini* with *high* reasoning setting and was also implemented using the OpenAI Agents SDK. It is scheduled to make an investment decision about a target stock once a month after the closing section of its first working day. Its investment decision is presented in a structured JSON format, consisting of a recommendation (buy, keep, or sell), a justification for the decision, and a target price. Similar to the workflow version of the financial agent, the LLM-based manager has at its disposal a *Python* interpreter, which can be used to compute any other metric that helps in the final decision.

As input, the manager receives a monthly time series covering the last 12 months of the target stock. Each month's information includes the price and fundamental indicators, as well as its previous recommendations. Price indicators include the opening, closing, average, minimum, and maximum prices of the stock on the first working day of the month, as well as the volume of buy/sell operations on that day. Fundamental indicators comprise the 32 computed by the proposed approach. Finally, past recommendations are provided together with their justifications and target prices for the target stock.

6.2 DATA

We evaluated the approach based on their monthly investment decisions from January 2024 to December 2025. We used a subset of 5 Brazilian publicly traded companies from the 13 used in the previous experiment: Alupar Investimento (ALUP11), Auren Energia (AURE3), Companhia Paranaense de Energia (CPLE3), Equatorial (EQTL3) and Eneva (ENEV3). Price information were extracted using the *FinBr* software[4]. To support financial analysts, the quarter- and annual reports of companies were extracted from *CVM* between September 2023 and September 2025.

6.3 BASELINES

To measure how well our fundamental analyst supports investment decisions, we conducted an ablation analysis by removing the fundamental indicators from the Financial Manager's input. We also use the *Buy-and-hold* strategy as a baseline.

6.4 EVALUATION

We evaluated the performance of the Financial Manager based on the sum of Trade Returns generated over the monthly decisions:

$$R_{\text{total}} = \sum_{s \in \mathcal{S}} \sum_{k=1}^{K_s} \frac{P_{s,k}^{\text{sell}} - \frac{1}{|B_{s,k}|} \sum_{p \in B_{s,k}} p}{\frac{1}{|B_{s,k}|} \sum_{p \in B_{s,k}} p} \tag{4}$$

[4] https://github.com/renanmoretto/finbr

where S is the set of stocks, K the set of trades, $P^{\text{sell}}_{s,k}$ is the closing price of stock s in the sell trading decision k and $\frac{1}{|B_{s,k}|}\sum_{p\in B_{s,k}} p$ is the average price of stock s across every buy decision before the selling one.

6.5 RESULTS

Table 3: Trade Return Sum of our proposed approach (*Full Approach*) and its baselines on 5 stocks between January 2024 and December 2025.

Model	ALUP11	AURE3	CPLE3	EQTL3	ENEV3	Total
Buy-and-hold	12.25%	-8.12%	46.87%	12.29%	54.63%	117.91%
Price-only	17.99%	24.34%	0.00%	0.00%	33.38%	75.71%
Full Approach	4.63%	18.75%	25.34%	20.08%	115.18%	183.98%

Table 3 depicts the total sum of the trade returns of each approach as well as per stock. Our approach (*Full Approach*) generates more returns than the baselines for 2 out of the 5 analyzed stocks: EQRL3 and ENEV3. The ablated version of our approach without the fundamental indicators leveraged the best returns for ALUP11 and AURE3, whereas the *Buy and Hold* decision gave the best returns for CPLE3. Overall, the LLM-based manager supported by the fundamental indicators provided by our fundamental analyst produces the highest total trade return, demonstrating the value of fundamental indicators for making good investment decisions and showing how well the proposed approach computes them.

7 DISCUSSION AND CONCLUSION

This study primarily investigates whether LLMs can perform traditional fundamental analysis of companies (RQ1). By modeling fundamental analysis as a task of computing 32 numerical indicators, we were able to automatically evaluate the proposed approach in the first experiment by comparing the generated indicators with reference indicators.

Results from this first experiment provide partial evidence in support of RQ1, showing that LLMs are capable of generating fundamental indicators with meaningful accuracy. Moreover, these results indicate that reasoning LLMs are more suitable for this task than non-reasoning ones (RQ2), and that model scale plays an important role, with the largest tested model (*mini*) achieving the best results (RQ3). In future work, larger versions (*GPT-4.1* and *GPT-5*) may be explored, as well as LLMs from other providers.

Still within the first experiment, we observed that the workflow architecture yielded better results than the agentic one, thus addressing RQ4. This finding highlights that granting full autonomy to LLMs when accessing databases introduces noise and degrades performance. This issue may be further investigated in future work aimed at selecting better models or optimizing prompts and tool descriptions toward a proper agentic solution for fundamental analysis. Finally, regarding RQ5, the proposed reflection mechanism appears useful for non-reasoning models, although it does not enable them to surpass reasoning models. Overall, the approach that generated fundamental indicators closest to the gold-standard values was our largest reasoning LLM (*gpt-5-mini*), implemented as a workflow without a reflection mechanism.

After identifying the best version of our fundamental analyst across all tested scenarios, we aimed to assess whether its generated outputs could support better investment decisions in financial trading simulations, thus further addressing RQ1 from a practical perspective. To this end, inspired by state-of-the-art work in financial trading using LLMs, we conducted a second experiment in which the fundamental indicators generated by our proposed financial agent were fed into an LLM-based financial manager that made buy/sell/hold decisions for Brazilian stocks. Compared with the *buy-and-hold* strategy and an ablated version in which the manager does not receive fundamental indicators, the approach that uses automatically generated fundamental indicators achieved the highest total trade return.

In conclusion, this study provides strong evidence that LLM-based solutions for fundamental company analysis are promising. Moreover, it paves the way for further Generative AI research in financial trading focused on markets beyond the US.

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A Appendix

A.1 Prompts

Figure 2 depicts the English-translated system prompt of our fundamental analyst, whereas Figure 3 shows the one from the financial manager.

A.2 Database Tools

Financial Reports Database Tool Figure 4 depicts the English-translated description of the financial reports database.

Stock Composition Database Tool Figure 5 depicts the description of the database tool from the Agent analyst to access information about the stock composition of the companies.

A.3 STRUCTURED OUTPUT SCHEMA

Figure 6 depicts the JSON schema used as structured output by the fundamental analyst and Figure 7 the one used by the financial manager.

Fundamental Analyst Prompt

You are a market analyst responsible for performing **fundamental analysis** of publicly traded companies in order to extract and compute relevant financial indicators, including but not limited to:

- **Total Assets**: Total assets represent the sum of all assets, rights, and resources owned by the company.
- **Cash and Cash Equivalents**: Cash and equivalents held by the company, including cash, bank balances, and highly liquid investments.
- **Current Assets**: Assets expected to be converted into cash in the short term.
- **Gross Debt**: Total outstanding debt, including short- and long-term debt and debentures.
- **Net Debt**: Gross debt minus cash and cash equivalents.
- **Equity**: Net equity owned by shareholders and controlling interests.
- **Net Revenue**: Total revenue minus taxes, discounts, and returns.
- **EBIT**: Earnings Before Interest and Taxes, approximating operating profit. Formula: Gross Profit – Selling Expenses – Administrative Expenses.
- **Net Income**: Profit after all expenses, interest, and taxes.
- **P/E (Price-to-Earnings)**: Share price divided by earnings per share.
- **P/B (Price-to-Book)**: Share price divided by book value per share.
- **P/EBIT**: Share price divided by EBIT.
- **PSR (Price-to-Sales)**: Share price divided by net revenue per share.
- **P/Assets**: Share price divided by total assets per share.
- **P/Working Capital**: Share price divided by working capital per share (Current Assets – Current Liabilities).
- **P/Net Current Assets**: Share price divided by net current assets per share.
- **EV/EBITDA**: Enterprise value divided by EBITDA (Operating Income + Depreciation + Amortization).
- **EV/EBIT**: Enterprise value divided by EBIT.
- **EPS (Earnings Per Share)**: Net income divided by the total number of shares.
- **BVPS (Book Value Per Share)**: Equity divided by the total number of shares.
- **Gross Margin**: Gross profit divided by net revenue.
- **EBIT Margin**: EBIT divided by net revenue.
- **Net Margin**: Net income divided by net revenue.
- **EBIT / Assets**: EBIT divided by total assets.
- **ROIC**: Return on invested capital, calculated as EBIT divided by (Total Assets – Trade Payables – Cash).
- **ROE**: Return on equity, calculated as Net Income divided by Equity.
- **Current Ratio**: Net current assets divided by current liabilities.
- **Gross Debt / Equity**: Gross debt divided by equity.
- **Asset Turnover**: Net revenue divided by total assets.

Guidelines:

- Use the Python interpreter to compute values whenever necessary.
- If the requested information is unavailable or cannot be derived, respond with `N/A`.
- Indicators such as Gross Margin, EBIT Margin, Net Margin, EBIT / Assets, ROIC, and ROE **must be reported as percentages with two decimal places (XX.XX%)**.
- To compute quarterly indicators (3 months), use the difference relative to the previous quarter.

Figure 2: Prompt used to guide the Fundamental Analyst agent in computing and interpreting financial indicators.

Investment Manager Prompt

You act as a market manager responsible for analyzing corporate financial information in order to support investment decisions.

Task Objective

Given the provided stock information, you must:

1. Make an investment decision on the requested date.
2. Present a justification in the form of a financial report.
3. Provide a target selling price.

Available Tools

You have access to a Python interpreter, which may be used to execute scripts, perform calculations, and extract any additional information required to support the final decision.

Input Data

You will receive a spreadsheet containing company-level information organized as follows:

Stock Identification

- `STOCK`: Stock ticker symbol.
- `TRADING_DATE`: Trading date under analysis.
- `LAST_PROCESSED_BALANCE_DATE`: Date of the most recently processed financial statement.

Previous Decision Information

- `PREVIOUS_JUSTIFICATION`: Justification used in the previous decision.
- `PREVIOUS_TARGET_PRICE`: Previously defined target price.
- `PREVIOUS_RECOMMENDATION`: Previous recommendation (e.g., buy, hold, or sell).

Price Information

- `OPEN_PRICE`: Opening price of the stock.

...

- `TRADED_QUANTITY`: Number of trades executed during the day.
- `TOTAL_TRADED_VOLUME`: Total number of shares traded during the day.

Fundamental Information

Accounting Data

- **Total Assets**: Total assets, rights, and resources owned by the company.

...

Valuation Indicators

- **P/E**: Price-to-earnings ratio.

...

Per-Share Indicators

- **EPS**: Earnings per share.
- **BVPS**: Book value per share.

Profitability and Margin Indicators

- **Gross Margin**: Gross profit divided by net revenue.

...

Capital Structure and Efficiency Indicators

- **Current Ratio**: Net current assets divided by current liabilities.

...

Guidelines

- Use the Python interpreter as a calculator whenever necessary.

Figure 3: Prompt used to guide the Investment Manager agent in making stock investment decisions based on market, price, and fundamental data.

SQL Tool Description

Query a SQL database containing **Standardized Financial Statements (DFP)** and **Quarterly Financial Information Reports (ITR)** from the Brazilian Stock Market, with the following schema:

```
CREATE TABLE IF NOT EXISTS DFP_ITR_CVM (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    CNPJ TEXT NOT NULL,                              -- Company tax ID
        (format: 00.000.000/0000-00)
    REPORT_DATE DATETIME NOT NULL,                   -- Report date
    COMPANY_NAME TEXT NOT NULL,                      -- Company name
    CVM_CODE TEXT NOT NULL,                          -- CVM identifier
        code
    DFP_GROUP TEXT NOT NULL,                         -- DFP statement
        group
    VERSION TEXT NOT NULL,                           -- Report version
    CURRENCY TEXT NOT NULL,                          -- Reporting
        currency
    EXERC_ORDER TEXT NOT NULL,                       -- Fiscal period
        order (Penultimate or Last)
    ANALYSIS_START_PERIOD_DATE DATETIME,             -- Analysis period
         start date
    ANALYSIS_END_PERIOD_DATE DATETIME NOT NULL,      -- Analysis period
         end date
    ACCOUNT_NUMBER TEXT NOT NULL,                    -- Financial
        statement account number
    ACCOUNT_NAME TEXT NOT NULL,                      -- Financial
        statement account name
    ACCOUNT_VALUE FLOAT NOT NULL,                    -- Account value
    IS_FIXED_ACCOUNT TEXT                            -- Indicates
        whether the account is fixed
);
```

Notes:

- The ACCOUNT_NUMBER encodes the hierarchical structure and ordering of balance sheet accounts. This information should be considered, together with dates, when ordering reports.
- Financial data are reported on a quarterly basis (March, June, September, and December), according to the analysis period start date.
- For annual financial statements, December reports should be used.

Figure 4: Schema used by the database tool from the Agent analyst to access standardized financial statements (DFP) and quarterly reports (ITR) from the Brazilian stock market.

SQL Tool Description

Query a SQL database containing **Share Capital Composition Reports** of companies listed in the Brazilian Stock Market, with the following schema:

```
CREATE TABLE IF NOT EXISTS CVM_SHARE_COMPOSITION (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    CNPJ TEXT NOT NULL,                         -- Company tax ID
        (format: 00.000.000/0000-00)
    REPORT_DATE DATETIME NOT NULL,              -- Report date
    COMPANY_NAME TEXT NOT NULL,                 -- Company name
    VERSION TEXT NOT NULL,                      -- Report version
    ORDINARY_SHARES_ISSUED INTEGER NOT NULL,    -- Issued common
        shares
    ORDINARY_SHARES_TREASURY INTEGER NOT NULL,  -- Treasury common
         shares
    PREFERRED_SHARES_ISSUED INTEGER NOT NULL,   -- Issued
        preferred shares
    PREFERRED_SHARES_TREASURY INTEGER NOT NULL, -- Treasury
        preferred shares
    TOTAL_SHARES_ISSUED INTEGER NOT NULL,       -- Total issued
        shares
    TOTAL_SHARES_TREASURY INTEGER NOT NULL      -- Total treasury
        shares
);
```

Notes:

- Share composition data are disclosed on a quarterly basis (March, June, September, and December), according to the report date.
- For annual share composition analysis, December reports should be used.

Figure 5: Schema used by the SQL tool to access share capital composition data of companies listed in the Brazilian stock market.

Indicator Output JSON Schema


```
{
  "$defs": {
    "Indicator": {
      "type": "string",
      "title": "Indicator",
      "enum": [
        "Asset",
        "Cash and Cash Equivalents",
        "...",
        "Asset Turnover"
      ]
    },
    "IndicatorOutput": {
      "type": "object",
      "title": "IndicatorOutput",
      "properties": {
        "indicator": {
          "$ref": "#/$defs/Indicator",
          "description": "Financial indicator name"
        },
        "value": {
          "type": "number",
          "description": "Indicator numeric value"
        }
      },
      "required": ["indicator", "value"]
    }
  },
  "type": "object",
  "title": "IndicatorOutput",
  "properties": {
    "indicators": {
      "type": "array",
      "title": "Indicators",
      "description": "List of computed financial indicators",
      "items": {
        "$ref": "#/$defs/IndicatorOutput"
      }
    }
  },
  "required": ["indicators"]
}
```


Figure 6: JSON schema used to represent the structured output of financial indicators computed by the financial analyst.

Financial Recommendation JSON Schema


```json
{
  "$defs": {
    "FinanceRecommendation": {
      "type": "string",
      "title": "FinanceRecommendation",
      "enum": [
        "Buy",
        "Sell",
        "Hold"
      ]
    }
  },
  "type": "object",
  "title": "FinanceOutput",
  "properties": {
    "recommendation": {
      "$ref": "#/$defs/FinanceRecommendation",
      "description": "Investment recommendation"
    },
    "justification": {
      "type": "string",
      "title": "Justification",
      "description": "Justification for the investment
          recommendation"
    },
    "target_price": {
      "type": "number",
      "title": "Target Price",
      "description": "Target selling price"
    }
  },
  "required": [
    "recommendation",
    "justification",
    "target_price"
  ]
}
```


Figure 7: JSON schema used to represent structured investment recommendations and target prices generated by the financial manager.